



Linux File Ownership – Practical Notes

1. Every file has 3 identity attributes

When you run:

```
ls -l
```

You see:

```
-rw-r--r-- 1 owner group file.txt
```

Linux tracks **three separate things**:

1. **Owner (User)**
2. **Group**
3. **Permissions (rwx)**

These are **independent** concepts.

2. Owner (User)

What the owner controls

The owner:

- automatically gets the **user permissions** (rwx)
- can change file permissions using `chmod`
- can change group ownership (with limits)
- does **NOT** automatically get delete rights

Owner $\neq$ delete authority

Deletion depends on directory permissions.

3. Group

What group ownership means

Group ownership defines:

- who gets the **group permission bits**
- shared access for teams
- collaboration control in directories

Example:

```
-rw-r----- alice devs file.txt
```

- alice $\rightarrow$ full control
- devs $\rightarrow$ read only
- others $\rightarrow$ no access

4. Permissions are split into 3 parts

```
rwx | rwx | rwx
```

```
user group others
```

Key idea:

- File permissions $\rightarrow$ control **file content**
- Directory permissions $\rightarrow$ control **file management**

5. Directory permissions (CRITICAL concept)

Permission	On files	On directories
r	Read file content	List filenames
w	Modify content	Create / delete / rename files
x	Execute file	Enter directory

👉 Delete needs w + x on directory, not on file.

6. chgrp – Change Group Ownership

What it does

Changes **only the group field**.

```
chgrp developers file.txt
```

Before:

```
alice users file.txt
```

After:

```
alice developers file.txt
```

Important rules

- You must be:
 - the file owner
 - AND a member of the target group
- Root can change anything

Recursive behavior

```
chgrp -R root project/
```

This changes:

- files ✓
- subdirectories ✓
- **project directory itself** ✓

This often removes write access unintentionally.

7. chown – Change Owner (and Group)

What it does

Can change:

- owner
- group
- both together

```
chown alice file.txt
```

```
chown alice:developers file.txt
```

```
chown :developers file.txt # group only
```

Power level

- Only **root** can change file owner
- Normal users can't give ownership to others

Recursive ownership change

`chown -R alice:developers project/`

Used for:

- fixing broken deployments
- moving files between users
- correcting copied data

8. Ownership vs Deletion (most misunderstood part)

You **can delete** a file even if:

- you don't own it
- group is root
- file is read-only

As long as:

You have write + execute permission on the directory

You **cannot delete** a file you own if:

- directory has no write permission